XXX **XXX**

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Research Experience

Search for low-MET electroweak supersymmetry

Texas Tech University

2022-ongoing

- Main topic of the thesis: This analysis focusses on all-hadronic signatures of stealth and RPV decays of charginos and neutralino of the supersymmetry
- The analysis will use CMS collision data from Run 2 (year 2016-2018) and Run 3 (year 2022-onwards)
- These extremely hard to probe models gain sensitivity using novel deep learning jet taggers, increased luminosity and better understanding of the detector
- Events with high energy gluon jets and top pairs are the dominant standard model backgrounds. A combination of particlenet bb, W jet taggers is used to improve sensitivity
- This analysis searches for supersymmetry in previously unexplored region using cutting edge AI and machine learning techniques. If we succeed in observing SUSY signatures, it would represent a major breakthrough in particle physics. If we fail to find any signatures, we will be able to place stronger upper limits on its production cross section.

Jnana Prabodhini Foundation

Pune, India & Lubbock, USA

RESEARCH COORDINATOR

2021

- A 3-year long project aimed at estimating COVID-19 mortality and undercount factor for Pune city in India.
- A frugal method of estimating COVID-19 mortality was tested against traditional statistics-based models widely used in epidemiology and media reports
- The data was collected through public surveying in 2 rounds over a period of a year
- The results were published in nature - scientific reports
- Gained experience in leading a team of researchers and working on topic outside the domain expertise.

Technical/Service Work

Hadronic Calorimeter (HCAL)

CERN, Geneva

2024

- Member of two international groups responsible for monitoring HCAL operations and performance studies
- Identified and mitigated HCAL operational anomalies: Diagnosed high-frequency "lumi spikes" in the forward HCAL and investigated anomalous peaks in total collected charge to ensure data quality for Run 3.
- Managed detector stability via noise monitoring : Developed and maintained the daily pedestal (noise) trend monitoring framework, directly providing the calibration constants required for downstream event reconstruction.

Light Dark Matter Experiment (LDMX)

CERN, Geneva

TEST BEAM PARTICIPANT

2022

- Developed slow control software for the prototype trigger scintillator for the LDMX
- Helped Construct a cosmic ray test stand at Texas Tech University, recorded and analyzed data
- Participated in test beam at CERN, where different detector components of LDMX were tested using high energy pion, electron and muon beams
- Monitored and controlled collision runs in shifts.
- Assumed charge of one of the subdetectors called trigger scintillator

Teaching Experience

Error analysis

Texas Tech University

LECTURER

Aug. 2023 - Nov. 2023

- Delivered lectures on error analysis required in conducting physics experiments
- The class composed of physics and maths majors
- Designed curriculum, weekly asignments and exams
- The lectures comprised of topics on statistics relevent to physics such as error propagation, Chi-square fitting, covariance etc.

Student guidance

2021 - 2023

- Mentored a student at Stanford University, who took over my work on LDMX trigger scintillator slow control software
- The work was included in the student's Masters thesis
- Mentored another student at Texas Tech University, who analyzed the data recorded at LDMX test beam
- Taught them how the data was recorded, program to analyze the data, features seen in the data
- The student presented the work at meeting of Texas-section of American Physical Society in 2023

Publications

Significant contribution to other areas

- "Conventional and frugal methods of estimating COVID-19-related excess deaths and undercount factors," Dedhe, A., Chowkase, A., Gogate, N., *et al.*, Sci Rep **14**, 10378 (2024) doi: 10.1038/s41598-024-57634-6
- "Current Status and Future Prospects for the Light Dark Matter eXperiment" T. Åkesson, N. Blinov, L. Brand-Baugher, C. Bravo, L. K. Bryngemark, P. Butti, C. Doglioni, C. Dukes, V. Dutta and B. Echenard, *et al.* arXiv:2203.08192

Author on 200+ CMS publications (selected list below)

- "Reweighting simulated events using machine-learning techniques in the CMS experiment", Eur. Phys. J. C **85**, 495 (2025), doi: 10.1140/epjc/s10052-025-14097-x
- "Development of systematic uncertainty-aware neural network trainings for binned-likelihood analyses at the LHC," Eur. Phys. J. C **85**, 1360 (2025) doi:10.1140/epjc/s10052-025-14713-w
- "Determination of the spin and parity of all-charm tetraquarks," Nature **648**, 58-63 (2025) doi:10.1038/s41586-025-09711-7
- "Searches for Higgs boson production through decays of heavy resonances," Phys. Rept. **1115**, 368-447 (2025) doi:10.1016/j.physrep.2024.09.004
- "The CMS Statistical Analysis and Combination Tool: Combine," Comput. Softw. Big Sci. **8**, 19 (2024) doi:10.1007/s41781-024-00121-4
- "Search for Soft Unclustered Energy Patterns in Proton-Proton Collisions at 13 TeV," Phys. Rev. Lett. **133**, 191902 (2024) doi:10.1103/PhysRevLett.133.191902

Workshops and conferences

2026	**Speaker**, APS global physics summit	*Denver, USA*
2024	**Participant**, European School of High Energy Physics	*Peebles, UK*
2024	**Speaker**, APS April Meeting	*Sacramento, USA*
2021	**Speaker**, Meeting of the Division of Particles and Fields of the American Physical Society	*Virtual*
2021	**Speaker**, Joint Meeting of the Texas Sections of APS, AAPT and Zone 13 of the SPS	*Virtual*

Education

Texas Tech University — *Lubbock, USA*

PhD in Physics — *Aug. 2020 - Ongoing*

- Currently pursuing PhD, with focus on exerimental high energy physics
- Supervisor: Prof. Sung-won Lee

Texas Tech University — *Lubbock, USA*

MS in Physics — *Aug. 2021*

- Masters degree offerred for clearing the PhD candidacy test

Indian Institute of Science Education and Research — *Pune, India*

BS-MS in Physics — *Aug. 2015 - May 2020*

- Masters thesis topic: To study charged pion performance in prototype HGCAL detector
- Masters thesis advisor: Prof. Seema Sherma

Honors & Awards

2026	**Recipient**, Bucy Graduate Scholarship in Applied Physics
2025	**Recipient**, Breakthrough Prize in Fundamental Physics
2024	**Recipient**, Bucy Graduate Scholarship in Applied Physics
2023	**Recipient**, Bucy Graduate Scholarship in Applied Physics
2022	**First prize**, Departmental poster presentation competition
2022	**Recipient**, Bucy Graduate Scholarship in Applied Physics
2021	**Recipient**, Departmental Research Associate Fellowship
2020	**Recipient**, Gogate foundation fellowship